

Project Proposal

Predicting Emergency Room Wait Times Using Machine Learning

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Abstract

This project aims to predict patient wait times in hospital emergency rooms using machine learning techniques. Accurate predictions can help hospitals manage resources efficiently, reduce patient overcrowding, and improve overall patient satisfaction.

Main Proposal Content

0.1 Motivation

Emergency Departments (EDs) are among the busiest hospital units, where patients require urgent care. Prolonged wait times increase treatment delays, patient risk, and reduce quality of care. Although prior studies have attempted to predict ED wait times, they often relied on limited features. ED waits result from complex interactions of patient demand, clinical acuity, operational capacity, staffing, and temporal crowding. This project proposes a machine learning approach to better capture these dynamics, reduce false-negative predictions of long waits, and provide actionable insights for operational management.

0.2 Datasets

This project utilizes three publicly available Emergency Room (ER) datasets from Kaggle to support a progressive modeling strategy.

0.2.1 Dataset 1: Hospital Emergency Room Dataset (Primary Dataset)

The primary dataset contains 9,216 emergency room visit records with 12 structured features per patient encounter. Variables include demographic attributes (age, gender, race), admission details, department referral, satisfaction score, and recorded waiting time. The target variable for our regression task is *patient_waittime*. This dataset represents patient-level clinical and operational information and serves as the foundation for baseline modeling. Dataset link: [Hospital Emergency Room Dataset](#).

0.2.2 Dataset 2: Simulated ER Wait Time Dataset (DMAIC Project)

The second dataset is a simulated emergency room dataset developed within a Lean Six Sigma (DMAIC) framework. It captures operational process variables related to ER flow and includes both pre-intervention and post-intervention data. This dataset allows exploration of how process optimization and system-level changes influence waiting times. Dataset link: [Simulated ER Wait Time Dataset](#).

0.2.3 Dataset 3: ER Wait Time Dataset (Rivalytics)

The third dataset provides simulated ER visit records including wait times and operational indicators. It introduces alternative structural representations of emergency care workflows, which can help test model robustness under different data distributions. Dataset link: [ER Wait Time Dataset](#).

Because these datasets differ in structure, feature representation, and collection methodology (realistic structured data versus simulation-based operational data), direct concatenation is not feasible. Instead, separate models will be trained on each dataset, and their predictions may be integrated using a stacked regression framework in Phase 2 of the project.

0.3 Related Work

Pak et al. (2021) employed machine learning regression techniques, including quantile regression and tree-based methods, to predict patient wait times in emergency departments. By leveraging queueing and service flow variables, they demonstrated that regression-based models substantially outperform traditional rolling average estimators in terms of mean squared prediction error (MSPE) and mean absolute prediction error (MAPE). Furthermore, quantile regression effectively mitigates underprediction of long wait times, highlighting the advantage of modeling wait time as a continuous variable. The full article is available at: [Pak et al., 2021](#). Building on this approach, our study adopts a supervised regression framework to predict exact ER wait times, preserving the continuous nature of the target variable. This methodology enables more precise and fine-grained operational insights and supports data-driven resource allocation and patient flow management in emergency care settings.

0.4 Intended Experiments

The goal of this project is to predict emergency room (ER) patient wait time using machine learning regression models. We will take a step-by-step approach, starting with one dataset and increasing complexity only if performance is not good enough.

0.4.1 Phase 1: Modeling with Dataset 1 (Primary Dataset)

We will first train models using Dataset 1. This dataset includes patient demographics, admission details, and recorded wait time. The following steps will be taken:

- Remove irrelevant identifiers (e.g., names, patient IDs).
- Handle missing values and encode categorical variables.
- Conduct a feature importance analysis to identify the most influential predictors of ER wait time.
- Train multiple regression models:
 - Linear Regression (baseline)
 - Decision Tree
 - Random Forest
 - Gradient Boosting (e.g., XGBoost/LightGBM)
- Validation Strategy
 - 80/20 train-test split
 - k-fold cross-validation

This phase evaluates whether patient-level features alone are sufficient for accurate prediction.

0.4.2 Phase 2: Incorporating Additional Datasets (If Needed)

If Dataset 1 has high error or low R^2 , we will train separate models in Dataset 2 (operational features) and Dataset 3 (hospital features). Since these datasets have different structures and features, we cannot combine them directly. Therefore, we will use stacked regression to merge predictions from the separate models through a meta-learner. This will allow us to make integrated predictions without combining incompatible data.

0.5 Evaluation

Since wait time prediction is a regression problem, we will evaluate performance using MAE, RMSE, and R^2 score.